

Counting domino tilings of a $3 \times 2n$ rectangle

Two proofs of a classical recurrence

Abstract

Let $a(n)$ be the number of ways to tile a $3 \times 2n$ board with dominoes. All three independent counting methods agree for $0 \leq n \leq 5$; the transfer matrix and coupled recurrences agree for $0 \leq n \leq 16$. A recurrence guessed from 13 terms survives four held-out terms and an exhaustive check to $n = 60$, and is then proved in two independent ways: by a column decomposition and by an order bound plus a finite check. The result is classical. We make the local decomposition and the eight-state transfer matrix explicit, distinguishing finite numerical checks from the arguments that establish the recurrence for every board size.

1 Result

Theorem 1. $a(0) = 1$, $a(1) = 3$ and $a(n) = 4a(n-1) - a(n-2)$ for $n \geq 2$. Equivalently

$$a(n) = \frac{3 + \sqrt{3}}{6} (2 + \sqrt{3})^n + \frac{3 - \sqrt{3}}{6} (2 - \sqrt{3})^n, \quad \frac{a(n+1)}{a(n)} \rightarrow 2 + \sqrt{3}.$$

The sequence is classical: $a(n)$ is term $n + 1$ of OEIS A001835 [1], whose indexing starts with 1, 1, 3, 11, 41, $\dots$. The proofs below reconstruct this known count and make no novelty claim.

2 How it was found

Three independent counts. A backtracking program fills the first free cell of a board, a transfer matrix over the $2^3 = 8$ column profiles counts tilings by matrix powers, and the coupled recurrences of Section 3 count them by induction. Backtracking and the transfer matrix agree for $0 \leq n \leq 5$; the transfer matrix and the recurrences agree for $0 \leq n \leq 16$. Boards with one corner removed agree between backtracking and the recurrence for $1 \leq m \leq 8$.

Guess, then test. From $a(0), \dots, a(12)$ an exact search finds a recurrence of order 2 with coefficients $(4, -1)$: eleven equations, two unknowns. No polynomial of degree ≤ 8 fits these discovery terms. The exponential growth established by Theorem 1, rather than the numerical plot in Figure 1 alone, rules out a polynomial formula of any fixed degree. Held out from the fit, $a(13), \dots, a(16)$ all satisfy the recurrence, and an exhaustive search over $2 \leq n \leq 60$ finds no counterexample.

n	$a(n)$	$4a(n-1) - a(n-2)$
0	1	
1	3	
2	11	11
3	41	41
4	153	153
5	571	571
6	2 131	2 131
7	7 953	7 953
8	29 681	29 681
9	110 771	110 771
10	413 403	413 403

Table 1. Exact counts and the recurrence.

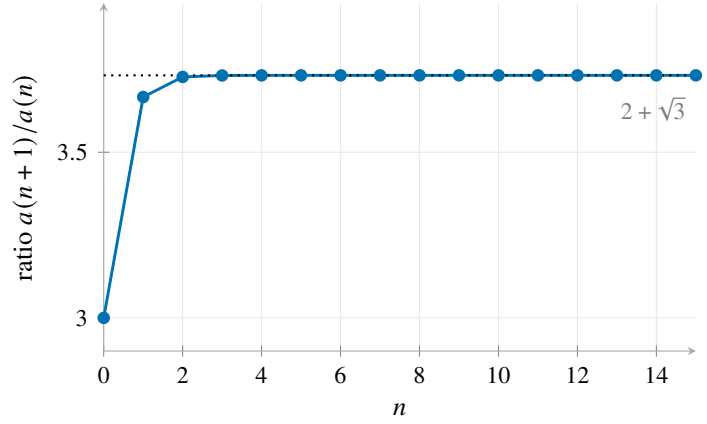


Figure 1. Consecutive-term ratios and the proved limit $2 + \sqrt{3}$.

3 Proof A: decomposing the board

Let A_m be the number of tilings of the $3 \times m$ board and B_m the number for the same board with one corner cell removed; set $A_0 = 1$, $A_1 = 0$, $B_0 = 0$, $B_1 = 1$.

$A_m = A_{m-2} + 2B_{m-1}$. Look at the three cells of the leftmost column. Either all three are covered by horizontal dominoes, which leaves a $3 \times (m-2)$ board; or a vertical domino covers the top two cells, which forces a horizontal domino on the bottom cell and leaves a $3 \times (m-1)$ board with its bottom-left corner removed; or the mirror image of that case. These cases are exhaustive and disjoint.

$B_m = A_{m-1} + B_{m-2}$ (corner cell $(3, 1)$ removed). The two remaining cells of the first column are covered by one vertical domino, leaving a full $3 \times (m-1)$ board; or by two horizontal dominoes, which forces a horizontal domino on the third cell of the second column and leaves a $3 \times (m-2)$ board with a corner removed.

Figure 2 shows the local configurations used in both identities. The grey cells form the residual board, and the crossed cell is absent.

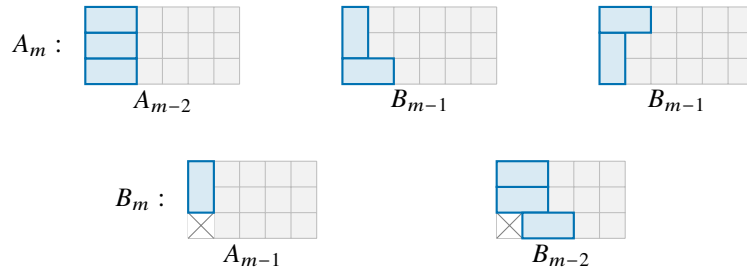


Figure 2. Exhaustive left-boundary cases. The top row shows a full board (drawn at $m = 6$); the bottom row shows a corner-removed board (drawn at $m = 5$). Coloured rectangles are forced dominoes, not additional choices. The residual counts indicated below each board give the two recurrences.

Eliminating B : $2B_{m-1} = A_m - A_{m-2}$ and $B_{m-1} = A_{m-2} + B_{m-3}$ give $A_m = 4A_{m-2} - A_{m-4}$ for $m \geq 4$. With $a(n) = A_{2n}$ this is Theorem 1. The closed form follows from the roots $2 \pm \sqrt{3}$ of $x^2 - 4x + 1$ and the values $a(0)$, $a(1)$, and the limit of the ratio from $|2 - \sqrt{3}| < 1$. $\square$

4 Proof B: an order bound and a finite check

A profile $s \in \{0, \dots, 7\}$ records cells already occupied by dominoes entering the current column. Bit r corresponds to row $r + 1$, counting from the top. The outgoing mask t records horizontal dominoes started in that column. Incoming and outgoing masks must be disjoint; the remaining cells must be empty or be one adjacent pair filled by a vertical domino. Thus $T_{s,t} = 1$ exactly when

$$s \& t = 0, \quad 7 \oplus (s | t) \in \{0, 3, 6\},$$

where the symbols denote bitwise AND, XOR, and OR. Otherwise $T_{s,t} = 0$. With rows and columns ordered $0, 1, \dots, 7$, this gives

$$T = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The matrix is symmetric. Starting and ending in profile zero forbids dominoes crossing either outer boundary, so a product of its entries counts each tiling exactly once. For example, $(T^2)_{00} = 3$, the three tilings of the 3×2 board.

The number of tilings of $3 \times m$ is $(T^m)_{00}$ for the 8×8 column-profile matrix T , so $a(n) = (T^{2n})_{00}$ is annihilated by a monic polynomial $P(E)$ of degree at most 8 in the shift E (Cayley–Hamilton for T^2). Let $Q(E) = E^2 - 4E + 1$ and $d = Q(E)a$. Polynomials in E commute, so $P(E)d = Q(E)P(E)a = 0$: the difference d obeys the same monic recurrence of order ≤ 8 , and a sequence like that vanishes once its first 8 terms do. Eight consecutive equations, $a(0), \dots, a(9)$, hold; hence $d \equiv 0$. This proof does not use Proof A and is the template implemented by `check_recurrence` with `order_bound=8`.

5 Confidence record

Statement	Evidence	Level
$a(n)$ for $n \leq 5$	backtracking, transfer matrix, recurrences agree	checked to $n = 5$, three methods
$a(n)$ for $n \leq 16$	transfer matrix and recurrences agree	checked to $n = 16$, two methods
$a(n) = 4a(n-1) - a(n-2)$ for all n	Proof A; Proof B; held-out terms; exhaustive to $n = 60$	proved (two independent proofs)
closed form and growth rate	algebra from the recurrence	proved
no polynomial formula	degree ≤ 8 search; exponential growth	proved by growth
novelty	OEIS A001835 lists the sequence	not claimed

Table 2. What each statement rests on, using the confidence ladder of the research-mode guide.

6 Generating function and comparison of the proofs

The recurrence gives a concise description as a formal power series. Put $F(x) = \sum_{n \geq 0} a(n)x^n$. Multiplying the recurrence by x^n and summing for $n \geq 2$ yields

$$(1 - 4x + x^2)F(x) = a(0) + (a(1) - 4a(0))x = 1 - x, \quad F(x) = \frac{1 - x}{1 - 4x + x^2}.$$

The two roots of the denominator recover the closed form in Theorem 1. No convergence assumption is needed for this formal-series identity.

Proof A uses two geometrically meaningful board families and explains the coefficient 4 by eliminating the corner-defect count. Proof B uses eight boundary profiles and makes the finite verification sufficient through a justified recurrence-order bound. The bound 8 is sufficient, not minimal; the actual order is 2. Agreement of numerical implementations is useful for finding errors, but the all-size conclusion rests on these structural arguments, not on the length of the numerical search.

7 Scope and reproducibility

This note is an exposition of a classical enumeration, not a claim of an original research contribution. The local cases, transition matrix, and finite-check argument can be examined independently. The archived checks document the specified finite ranges; no proof assistant verifies the entire manuscript or its runtime.

Reproduce. From the repository root, run

```
uv run --locked python demos/domino-research/reproduce/explore.py
```

This recomputes the numbers; `build_note.py` typesets this note from the archived record.

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A001835, oeis.org/A001835, accessed October 9, 2026.